

Forecasting U.S. Real GDP Growth with Shrinkage Methods and Diffusion Indices

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I. Abstract

This project forecasts U.S. real GDP growth using Shrinkage Methods and Diffusion Indices. I pull real GDP from FRED and compare six forecasting models in a pseudo-out-of-sample experiment from 2015 onward, which include the random walk, the random walk with drift, and three big-data models: a diffusion-index regression on the first three principal components, a Ridge regression, and a Lasso regression. Forecasts are made one quarter ahead and four quarters ahead. This project aims to find if, for a one-quarter horizon or a four-quarter horizon, the simple sample mean beats every other model, including all of the big-data methods.

II. Introduction

Gross Domestic Product, or GDP, is a broad measure of how much the economy produces. Real GDP adjusts for inflation, so changes in real GDP are a standard measure of whether the economy is expanding or slowing. Forecasting GDP growth is a difficult task because the economy is affected by many factors, such as employment, prices, interest rates, consumer spending, financial markets, housing, and capital investment. A simple forecast can often perform surprisingly well. Therefore, this project builds and compares six different models on the same data, in the same way, so they can be evaluated side by side. The simple models are the naive mean, the random walk, and the random walk with drift. The big-data models summarize many predictors with a few principal components, and two shrinkage regressions, Ridge and Lasso, that estimate a regression with many predictors while preventing the model from overfitting. Forecast accuracy is summarized by the mean squared forecast error (MSFE), the mean absolute error (MAE), and the MSFE relative to the random walk benchmark.

III. Methodology

(a) Data and Training Sample via FRED

The sample data starts in 1985, and the final common sample ends in 2025Q4. The training period ends in 2014Q4. Forecasts after that date are treated as pseudo out-of-sample forecasts.

(b) Evaluation Metrics

The random walk uses the most recent GDP-growth value. The drift model adds the average historical change. The diffusion-index model compresses the large predictor panel into the first few principal components and then runs an OLS regression. Ridge and Lasso use the full standardized predictor panel but shrink coefficients to reduce overfitting. Lasso can also set some coefficients exactly to zero. A relative MSFE below 1 means the model had a lower squared forecast error than the random walk. A value above 1 means it did worse than the random walk.

(c) Forecasting Models

$$\text{GDP Growth: } y_t = (\ln(\text{GDPC1}_t) - \ln(\text{GDPC1}_{t-1}))$$

$$\text{Naive mean: } \hat{y}_{T+h|T} = \left(\frac{1}{T}\right) \times \sum y_t = \bar{y}$$

$$\text{Random Walk: } \hat{y}_{T+h|T} = \bar{y} \quad \bar{y}$$

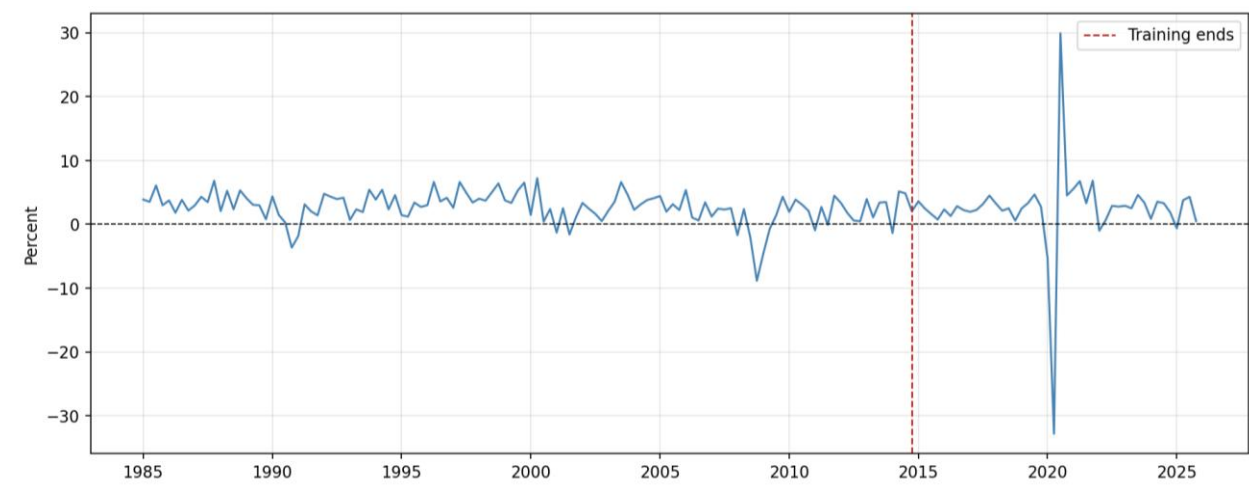
$$\text{Random Walk with Drift: } \hat{y}_{T+h|T} = y_T + h \times \delta_T$$

$$\text{Diffusion: } y_{t+h} = \beta_0 + \beta_1 F_{1,t} + \beta_2 F_{2,t} + \beta_3 F_{3,t} + \varepsilon_{t+h}$$

IV. Results

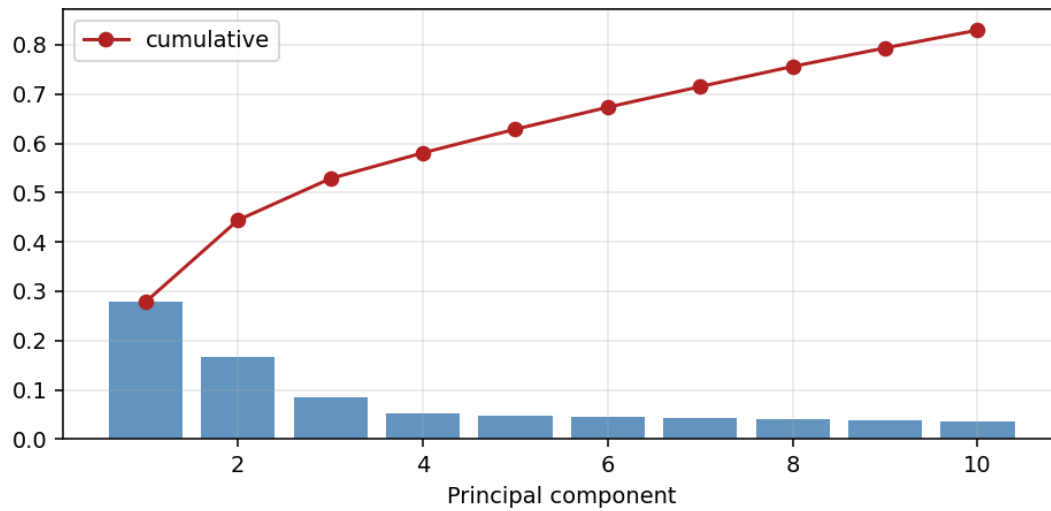
(a) Data Series

Quarterly real GDP growth has averaged about 0.64% per quarter over the sample, with a standard deviation of about 1.06. The series witnessed a record collapse during the COVID-19 pandemic in 2020, followed by a record rebound. Overall, growth fluctuates modestly around its average.



(b) Diffusion-index factors

The scree plot shows the share of the predictor panel's variance explained by each principal component. The first factor alone captures about 28% of the variance, the second adds about 16%, the third adds another 9%, the fourth and fifth both add about 5%, for a cumulative total of about 63% in the first five factors.



(c) Actual vs Forecast

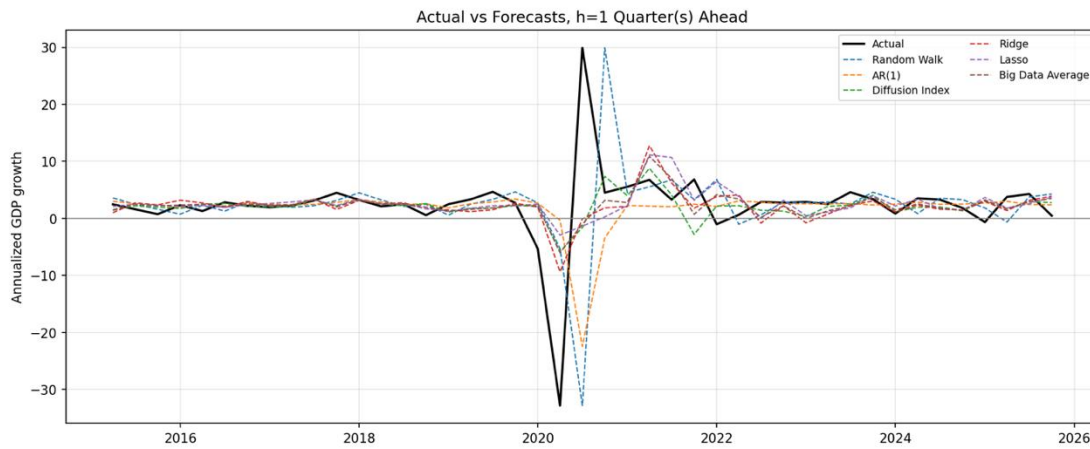


Figure 1: One-quarter-ahead forecasts.

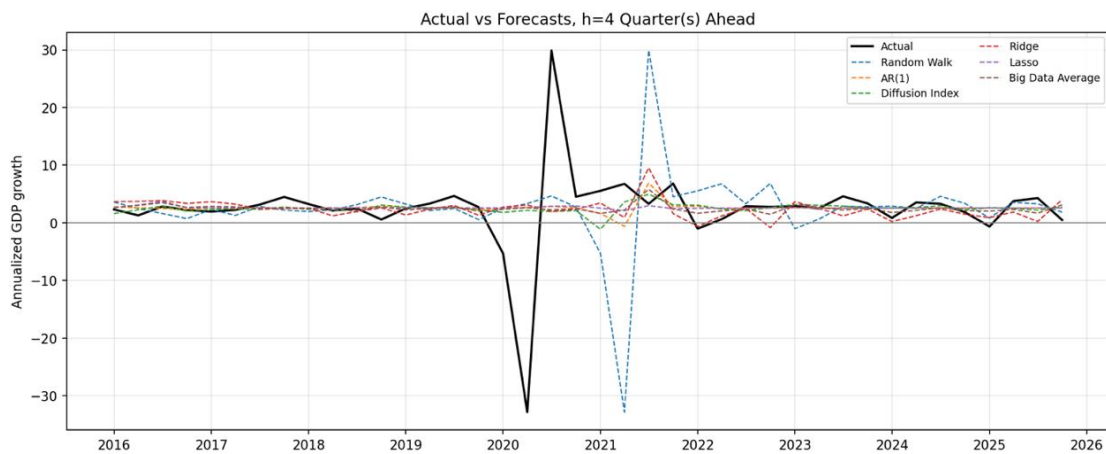


Figure 2: Four-quarter-ahead forecasts.

(d) Forecast table

Forecast error: $e_{t+h} = y_{t+h} - \widehat{y_{t+h}|t}$

$$MSFE = \frac{1}{n} \sum_{i=1}^n e_{t+h}^2$$

$$MAE = \frac{1}{n} \sum_{i=1}^n e_{t+h}$$

$$Rel\ MSFE = \frac{MSFE_{model}}{MSFE_{Random\ Walk}}$$

The table below shows MSFE, MAE, and relative MSFE versus the random walk for all six models, at both forecast horizons.

Horizon	Model	MSFE	MAE	Rel. MSFE vs RW
h=1	Naïve Mean	3.05	0.69	0.392
h=1	Random Walk	7.76	1.06	1.00
h=1	Random Walk with Rift	7.82	1.04	1.01
h=1	Diffusion Index	5.61	0.85	0.72
h=1	Ridge	7.88	0.95	1.02
h=1	Lasso	9.73	0.93	1.25
h=4	Naïve Mean	3.02	0.68	0.47
h=4	Random Walk	6.43	1.23	1.00
h=4	Random Walk with Rift	6.63	1.25	1.03
h=4	Diffusion Index	3.15	0.78	0.49
h=4	Ridge	3.03	0.69	0.47
h=4	Lasso	3.05	0.70	0.48

V. Discussion

The big-data models have an advantage because they can use information from many parts of the economy. Principal components summarize the common movement, while Ridge and Lasso try to use the individual predictors directly without letting the model become too unstable. At the same time, GDP growth is a difficult target. It is revised over time, measured quarterly, and can move sharply during recessions and recoveries. These features make simple benchmarks hard to beat consistently. One important consideration is the COVID-19 pandemic, which was observations of GDP growth in quarter 2 of 2020, which is a historically unprecedented collapse, and quarter 3 of 2020, which is a sharp rebound of comparable magnitude. Because MSFE squares forecast errors, even a single extremely large miss can dominate the entire out-of-sample

loss statistic. All six models, naive mean or big data, failed to anticipate the pandemic shock, which was by nature unpredictable from historical data patterns. A useful extension would be to recompute MSFE, excluding the four COVID quarters from quarter 1 of 2020 to quarter 4 of 2020, to determine whether the relative rankings of models change when these outliers are removed.

VI. Conclusion

The results show why the forecasting with simple benchmarks is strong, but larger macroeconomic information sets can help at some horizons. Big-data methods such as Ridge, Lasso, and a three-factor diffusion index beat the random-walk substantially at both horizons but failed to beat the simple expanding-window mean at one quarter ahead. At four quarters, four models, Naive Mean, Diffusion Index, Ridge, and Lasso, are essentially tied, each delivering roughly half the random walk's MSFE. Big-data methods are a meaningful improvement over the random walk but matching or beating the conditional mean of a stationary, mean reverting target like GDP growth is difficult, and the marginal value of additional predictors depends strongly on the forecast horizon and on the noise in the data.

VII. Reference

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